Kinetic Theory

Models of a Gas

- Hard Sphere Model
  1. particles of finite sizes
  2. collisions are the only interaction.

- Diameter of a container L
- Average separation S
- Radius of a molecule r_c
- Average distance between collisions λ

$$L \gg \lambda \gg S \gg r_c$$

Classical Limits

- Quantum limit
  $$\frac{N}{V}\lambda^3 \ll 1.$$
  λ is de Broglie wavelength
- Relativistic Limit
  $$k_B T = \frac{m}{2}v^2 \ll mc^2$$
  $$T \ll 10^{14}\ K$$

Pressure in a simple Model

$$P_x = \frac{N}{V} m \langle V_x^2 \rangle$$

$$P = \frac{N}{3V} m \langle V^2 \rangle \qquad P = \frac{n}{3} m \langle v^2 \rangle$$

- particles moving along a line
- 2 walls separated by L

Pressure & Energy Density

- translational kinetic energy
  $$\langle K \rangle = \frac{m}{2}\langle V^2 \rangle$$
  $$P = \frac{2}{3}\frac{N}{V}\langle K \rangle$$
- Energy density
  $$u = \frac{U}{V}.$$

$$P = \frac{2}{3}u_K.$$

Flux

$$\vec{j} = n\vec{v}$$

- Momentum Flux
  $$P_{x,i} = \iiint mnF\, V_x^2\, dv^3$$
  Assume $F(V_x \cdots) = F(-V_x \cdots)$
  $$P_x = 2P_{x,i} = mn\langle V_x^2 \rangle$$
- $F(V_x, V_y, V_z)$ is the distribution of velocity

Speed Distribution

$$\int_0^\infty f(v)\, dv = 1$$

Velocity Distribution

$$\iiint_{-\infty}^{\infty} F(V_x, V_y, V_z) = 1$$

- isotropic
  $$\langle V_x^2 \rangle = \langle V_y^2 \rangle = \langle V_z^2 \rangle = \frac{1}{3}\langle V^2 \rangle$$
  $$\langle V_x \rangle = \langle V_y \rangle = \langle V_z \rangle = 0$$

Velocity Distribution in Spherical Coordinate

$$v^2 \sin\theta\, dv\, d\theta\, d\varphi$$

- isotropic.
  $d\theta$ not matter
  $$f_{sp} = \frac{1}{4\pi} f(v) \sin\theta$$
- any distribution
  $$f_{sp} = v^2 \sin\theta\, F(V_x, V_y, V_z)$$

Effusion

- condition
  radius of a hole: a
  mean free path: λ
  $$a \ll \lambda$$
- Effusion Flux
  $$\text{rate} = n \cdot V \div t$$
  $$= n f(v)\, dv\, \frac{\sin\theta\, d\theta\, d\varphi}{4\pi}\, \frac{A v t \cos\theta}{t}$$

$$\Phi = \int \text{rate} = \frac{n}{4}\langle v \rangle$$

Effusion Rate

$$\frac{1}{4} A n \langle v \rangle$$

isotropic gas

Equilibrium

since $\langle v \rangle \propto \sqrt{T}$

$$n_1\sqrt{T_1} = n_2\sqrt{T_2}$$

Effused Particles (Speed)

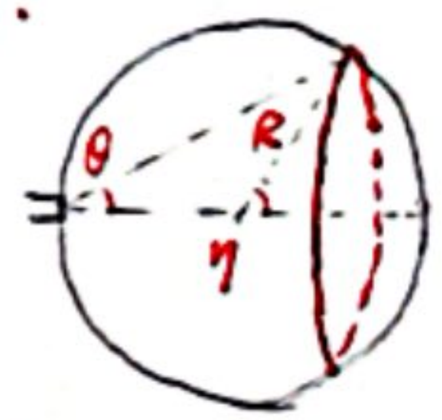
- particles effused from a hole has a speed distribution $\propto v f(v)$ if the distribution before effusion is $f(v)$.
- because for a given time t, the region contributing to effusion is $A \cdot vt\cos\theta$.

$$f \propto v^3 e^{-\alpha v^2}$$

$$\frac{m}{2}\langle V^2 \rangle = 2k_B T$$

Effused Particle (Angle)

- Effused particles hits a collecting sphere uniformly.
- # of particles in the range $\theta \to \theta + d\theta$
  $$N \sin\theta \cos\theta\, d\theta$$



- Area of "red strip"
  $$2\pi R \sin\eta \cdot d\eta.$$
- amount per area. $(\eta = 2\theta)$
  $$N/8\pi R \sim \text{constant}$$

Everything before Mean Free Path is done without intermolecular collisions.

Maxwell - Boltzmann Distribution

Speed pdf.

$$f(v) = A v^2 e^{-\frac{mv^2}{2k_BT}}$$

$$A = \sqrt{\frac{2}{\pi}} \left(\frac{m}{k_BT}\right)^{\frac{3}{2}}$$

Δ : standard deviation

$$\Delta = \sqrt{\frac{kT}{m}}$$

Other Properties

$$\langle v \rangle = \sqrt{\frac{8}{\pi}}\,\Delta$$

$$V_{rms} = \sqrt{3}\,\Delta$$

$$V_{peak} = \sqrt{2}\,\Delta$$

Temperature & Mean Translational Kinetic Energy

$$\bar{E} = \frac{m}{2}\langle v^2 \rangle = \frac{3}{2}kT$$

Equation of State

$$p = nkT$$

$$p = \frac{2}{3}u \,. \quad u = \frac{\bar{E}N}{V}$$

Boltzmann Factor

since particles are free to exchange energy, thermal equilibrium is attained when the distribution is exponential

State Space : v^2

- more state spaces at higher v
- # of state spaces $\propto$ volume
- in the range $v \to v + dv$

$$V = 4\pi m^3 v^2 dv$$

- here comes the factor v^2

Distribution of Each Component

- change between Cartesian and Spherical coordinates

$$f_x(v_x) = N e^{-\frac{1}{2}\frac{mv_x^2}{kT}}$$

$$N = \frac{1}{\sqrt{2\pi}}\frac{1}{\Delta}$$

Doppler Broadening

- if an atom is in motion at v relative to the lab.

$$(\nu - \nu_0) = \frac{v_x}{c}\nu_0$$

- Frequency Distribution

$$g(\nu) = f_x(v_x) \cdot \left|\frac{dv_x}{d\nu}\right|$$

$$g(\nu) = \frac{c}{\nu_0} N e^{-\frac{mc^2(\nu-\nu_0)^2}{2kT\nu_0^2}}$$

- standard deviation

$$\sigma_\nu = \frac{\nu_0}{c}\sqrt{\frac{kT}{m}}$$

Heat Capacity

- 2 principle heat capacities

$$C_V = \frac{\partial Q}{\partial T}\Big|_V. \quad C_p = \frac{\partial Q}{\partial T}\Big|_p$$

- ideal gas

$$C_p = C_V + Nk_B$$

- Monatomic

$$C_V = \frac{3}{2}Nk_B. \quad C_p = \frac{5}{2}Nk_B$$

Equipartition Theorem

- each quadratic independant term contributes to an energy of $\frac{1}{2}kT$.
- Monatomic

$$H = \sum_1 \frac{p_{xi}^2}{2m} + \frac{p_{yi}^2}{2m} + \frac{p_{zi}^2}{2m}$$

$$\langle E \rangle = \frac{3}{2}kT$$

- Diatomic

$$H = \sum_1 \frac{p_i^2}{2m} + \frac{L_x^2}{2I_x} + \frac{L_y^2}{2I_y} + \left(\frac{L_z^2}{2I_z}\right)_{bb}$$

$$\langle E \rangle = \frac{5}{2}kT$$

- Diatomic at higher T

1 vibrational mode $\to$ 2 quadratic terms

$$\langle E \rangle = \frac{7}{2}kT$$

- activate rotation $\Delta E = \frac{\hbar^2}{2I_i}$
- activate vibration $\Delta E = \hbar\omega$

$$k \cdot T \geq \Delta E$$

Mean Free Path

- collision cross section

$$A = \pi(r_1 + r_2)^2$$

- $\langle \lambda \rangle$
- for faster molecules

$$\lambda = \frac{L}{N} = \frac{1}{n\sigma}$$

- for slower molecules

$$\lambda = \frac{v}{\bar{v}}\frac{1}{\sigma n} \text{. from effusion}$$

$$\lambda = \frac{v}{\bar{v}}\frac{1}{\sigma n}$$

- Speed Averaged λ

$$\langle \lambda_v \rangle = 0.6775\frac{1}{n\sigma}$$

- Mean Free Path

$$\lambda_0 = \frac{1}{\sqrt{2}n\sigma}$$

Comparison of λ, $\langle \lambda_v \rangle$, λ_0

λ: mean free path of a molecule at a certain speed.

$\langle \lambda \rangle$: average mean free path over Maxwell-Boltzmann distribution

λ_0: standard value. obtained from relative speed and collision rate.

Distribution of Free Path Length (at a given v)

- # of molecules not collided

$$N = N_0 e^{-kx}. \quad k = \frac{1}{\lambda_v}$$

- the probability of no collision within distance x.

$$P(x) = e^{-kx}. \quad k = \frac{1}{\lambda_v}$$

- distribution function of λ_v

$$f(x)dx = P(x) - P(x+dx)$$

$$f(x) = \frac{1}{\lambda_v} e^{-\frac{x}{\lambda_v}}$$

Distribution of λ in a whole gas

$$g(x) = \int_0^\infty \frac{1}{\lambda_v} e^{-\frac{x}{\lambda_v}} f_v(v)dv$$

$$g(x) = 1.6 n\sigma e^{-1.6 n\sigma x}$$

Transport Equation

- continuity equation

$$\frac{\partial \rho}{\partial t} + \vec{\nabla}\vec{J} = 0$$

- Fick's law

number flux $\propto$ - gradient of n

$$\vec{J} = -D\vec{\nabla} n$$

- Fourier's law

heat flux $\propto$ - temperature gradient

$$\vec{J} = -K\vec{\nabla} T$$

- Transport Equation

$$\frac{\partial}{\partial t}\psi = c\nabla^2\psi$$

Before transport equation, everything is treated in equilibrium.

Viscosity (Momentum Transport)

- viscosity quantifies the transport of momentum

$$\frac{F_x}{A} = -\eta \frac{du_x}{dz}$$

- shear stress is in the direction of flow.
- z is perpendicular to the flow.
- kinetic theory estimate

$$\eta \simeq \frac{1}{3} n m \bar{v} \lambda_0$$

- independent of number density

$$\lambda \propto \frac{1}{n}.$$

no dependence on n

Thermal Conductivity (Energy transport)

- Fourier's law

$$\vec{J} = -K\vec{\nabla} T$$

- kinetic theory estimate

$$K \simeq \frac{1}{3} C_v \bar{v} \lambda_0$$

C_v: heat capacity per unit volume.

$$K \simeq \frac{1}{3} n C_1 \lambda \bar{v}$$

C_1: heat capacity per particle

- independent of number density

The models assume the mean free path λ_0 is much smaller than the dimensions of the container.

$\lambda_0 \ll L$

Green's Function.

- it helps to solve initial value problem
- if there is a particle at $\vec{x}$, where will it be later.

$$G(x,0) = \delta(x).$$

- For any initial density

$$n(\vec{x},t) = \int G(\vec{x'}-\vec{x},t)\, n(x',0)\, dx'$$

Solution in ℓ dimension

$$G(\vec{x},t) = \frac{1}{(4\pi Dt)^{\ell/2}} \exp\left(-\frac{|\vec{x}|^2}{4Dt}\right)$$

Sources & Sinks

$$\frac{\partial \rho}{\partial t} + \vec{\nabla}\vec{J} = S(\vec{r},t)$$

$S>0$. source

$S<0$. sink.

- Diffusion

$$\frac{\partial n}{\partial t} = D\nabla^2 n + S$$

- Heat Conduction

$$\frac{\partial T}{\partial t} = \frac{k}{c}\nabla^2 T + \frac{\dot{q}}{c}.$$

$\dot{q}$ is heating power per unit volume.